



AUTUMN END SEMESTER EXAMINATION-2023
3rd Semester B.Tech

AUTOMATA THEORY AND FORMAL LANGUAGES
CS21003

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 3 Hours

Full Marks: 50

Answer any SIX questions.

Question paper consists of four SECTIONS i.e. A, B, C and D.

Section A is compulsory.

Attempt minimum one question each from Sections B, C, D.

The figures in the margin indicate full marks.

All parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions:

[1 × 10]

- (a) Arrange the following in ascending order of their power of acceptance:
NPDA, DFA, NFA, DPDA, TM (the abbreviations have their usual meanings.)
- (b) Distinguish between NFA and DFA with suitable examples.
- (c) Design an NFA with no more than five states for the language $L = \{abab^n : n \geq 0\} \cup \{aba^n : n \geq 0\}$.
- (d) What is the importance of stack as an external memory in Push Down Automata?
- (e) *If L is a context free language, then its complement is also context free.*

The above statement is true or false? Justify.

- (f) Give examples of two context free languages whose intersection is not a CFL.
- (g) State whether the given CFG is ambiguous or not.
 $S \rightarrow a \mid abSb \mid aAb$
 $A \rightarrow bS \mid aAAb$
- (h) Design an NPDA equivalent to the CFG
 $A \rightarrow aBC \mid bD \mid a$
- (i) Give the formal definition of Turing Machine.
- (j) What do you mean by Chomsky Hierarchy? Give a Chomsky classification of languages accepted by different automata.

SECTION-B

2. (a) Find a Right Linear Grammar (RLG) for the language [4]
 $L = \{a^n b^m : n + m \text{ is odd}\}.$
 Convert the RLG to Left Linear Grammar (LLG).
- (b) Find minimal DFA for the following language over $\Sigma = \{a, b\}.$ [4]
 $L = \{a^n : n \bmod 3 = 1\} \cup \{a^n : n \bmod 5 = 1\}.$
3. (a) Construct a DFA that accepts the language generated by the following grammar. Write its regular expression. [4]
 $S \rightarrow abS \mid A$
 $A \rightarrow baB$
 $B \rightarrow aA \mid bb$
- (b) Design an NFA that accepts the language [4]
 $L (ab^*aa + bba^*ab).$
 Convert the NFA to minimal DFA.

SECTION-C

4. (a) Find context-free grammars for the following language: [4]

i) $L = \{w \in \{a, b\}^* : n_a(w) \neq n_b(w)\}$

ii) $L = \{a^n b^m c^k : k = |n-m|, n \geq 0, m \geq 0, k \geq 0\}.$

- (b) Write down the statement of Pumping Lemma for context free language. Prove that the following language is not context free: [4]

$$L = \{a^n : n \text{ is a perfect square}\}$$

5. (a) Construct an NPDA that accept the following language [4]
on $\Sigma = \{a, b, c\}$:

$$L = \{a^n b^{n+m} c^m : n \geq 0, m \geq 1\}.$$

- (b) Transform the following grammar into Chomsky Normal Form (CNF). [4]

$$S \rightarrow baAB,$$

$$A \rightarrow bAB|\lambda,$$

$$B \rightarrow BAa | A | \lambda$$

6. (a) Construct an NPDA that accept the following language [4]
on $\Sigma = \{a, b, c\}$.

$$L = \{a^i b^j c^{i-j} : i \geq j \text{ and } i, j \geq 1\}.$$

- (b) Convert the following grammar into Greibach Normal Form (GNF). [4]

$$S \rightarrow ABb | a | b$$

$$A \rightarrow aaA | B$$

$$B \rightarrow bAb$$

SECTION-D

7. Design a Turing machine for the following language on [8]
 $\{a, b, c\}$
 $L = \{a^n b^n c^n : n \geq 1\}$
Give a sequence of Instantaneous Descriptions (IDs) for the generated Turing machine leading to the acceptance of the string 'aabbcc'.
8. (a) Design a Turing machine for the following language on [4]
 $\{x, y\}$
 $L = \{w : |w| \text{ is a multiple of } 4\}$
- (b) Prove that context free languages are closed under [4]
union but not closed under intersection.
